TRIP COMPUTER INFORMATION SYSTEM

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Outline

- The trip computer system displays instantaneous fuel economy, average fuel economy (with tachometer), remaining travel distance, average vehicle speed (with tachometer), and vehicle speed alarm (vehicle speed alarm).

Function

- The instrument cluster controls the trip computer system based on the following signals.

Signal name	Sending module/part name	Communication method
Vehicle speed signal	PCM	HS-CAN
Fuel injection amount signal		
Traveled distance signal		
Steering switch operation signal (with tachometer)	Start stop unit	

Instantaneous fuel economy calculation function

- The instrument cluster calculates the instantaneous fuel economy based on the fuel injection amount and traveled distance over the past **2 s** when the vehicle speed is at the set value or more. In addition, the calculation results are updated **every 2 s**.

Instantaneous fuel economy formula

A TYPE

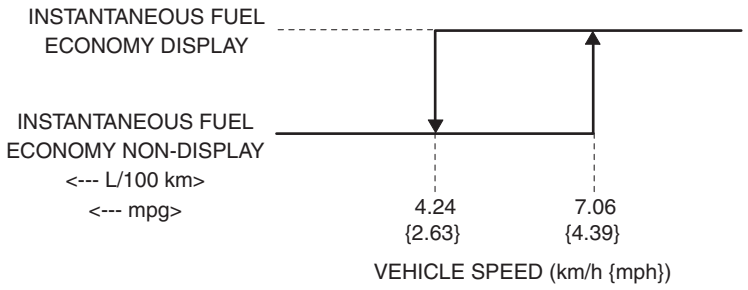
$$\text{INSTANTANEOUS FUEL ECONOMY (L/100 km)} = \frac{\text{FUEL CONSUMPTION (L) OVER PAST 2 s X 100}}{\text{TRAVELED DISTANCE (km) OVER PAST 2 s}}$$

B TYPE

$$\text{INSTANTANEOUS FUEL ECONOMY (mpg)} = \frac{\text{TRAVELED DISTANCE (mile) OVER PAST 2 s}}{\text{FUEL CONSUMPTION (gallon) OVER PAST 2 s}}$$

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- When there is no fuel consumption, such as during engine braking, <0.0 L/100 km> or <99.9 mpg> is displayed.
- The instantaneous fuel economy is displayed when the vehicle speed reaches **7.06 km/h {4.39 mph} or more**, and <--- L/100 km> or <--- mpg> is displayed when the vehicle speed decreases to **4.24 km/h {2.63 mph} or less**.



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Average fuel economy calculation function (With tachometer)

- The instrument cluster calculates the average fuel economy based on the traveled distance and fuel consumption from connecting the battery or resetting the average fuel economy. In addition, the calculation results are updated every minute.

Average fuel economy formula

A TYPE

$$\text{AVERAGE FUEL ECONOMY (L/100 km)} = \frac{\text{CUMULATIVE FUEL CONSUMPTION}^{*1}\text{ (L) X 100}}{\text{CUMULATIVE TRAVELED DISTANCE}^{*2}\text{ (km)}}$$

B TYPE

$$\text{AVERAGE FUEL ECONOMY (mpg)} = \frac{\text{CUMULATIVE TRAVELED DISTANCE}^{*2}\text{ (mile)}}{\text{CUMULATIVE FUEL CONSUMPTION}^{*1}\text{ (gallon)}}$$

*1: SUM OF FUEL CONSUMPTION FROM CONNECTING BATTERY OR RESETTING AVERAGE FUEL ECONOMY TO PRESENT

*2: SUM OF TRAVELED DISTANCE FROM CONNECTING BATTERY OR RESETTING AVERAGE FUEL ECONOMY TO PRESENT

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- When the cumulative traveled distance is **0 km {0 mile}**, <99.9 L/100 km> or <0.0 mpg> is displayed.
- If the INFO button is pressed for **1.5 s or more** while the average fuel economy is displayed, the cumulative fuel economy and cumulative traveled distance data are reset.
- If tripmeter A is reset, the average fuel economy is also reset.
- During the **1 min period** after the data has been reset, <--- L/100 km> or <--- mpg> is displayed.

Remaining travel distance calculation function

- The instrument cluster calculates the remaining travel distance based on the instantaneous fuel economy at **0.1 km {0.06 mile}** prior, cumulative fuel economy, and the fuel tank level. In addition, the calculation results are updated **every 1 s**.

Remaining travel distance formula

A TYPE

$$\text{REMAINING TRAVEL DISTANCE (km)} = \frac{\text{FUEL TANK LEVEL (L \{US gal, Imp gal\})}}{\text{CUMULATIVE FUEL ECONOMY}^{*1}\text{ (L/km) X (1500-1) / 1500 + INSTANTANEOUS FUEL ECONOMY (L/km) IN PAST 0.1 km INTERVAL / 1500}}$$

B TYPE

$$\text{REMAINING TRAVEL DISTANCE (mile)} = \frac{\text{FUEL TANK LEVEL (L \{US gal, Imp gal\})}}{\text{CUMULATIVE FUEL ECONOMY}^{*1}\text{ (L/mile) X (1500-1) / 1500 + INSTANTANEOUS FUEL ECONOMY (L/mile) IN PAST 0.06 mile INTERVAL / 1500}}$$

*1: SUM OF FUEL ECONOMY EVERY 0.1 km {0.06 mile} FROM CONNECTING BATTERY TO PRESENT

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- The instrument cluster displays <0 km> or <0 miles> when the fuel tank level is **6 L {2 US gal, 1 Imp gal}**.

Average vehicle speed calculation function (With tachometer)

- The instrument cluster calculates the average vehicle speed based on the cumulative traveled distance and cumulative traveled time from connecting the battery or resetting the average vehicle speed. In addition, the calculation results are updated **every 10 s**.

Average vehicle speed formula

A TYPE

$$\text{AVERAGE VEHICLE SPEED (km/h)} = \frac{\text{CUMULATIVE TRAVELED DISTANCE (km)}}{\text{CUMULATIVE TRAVELED TIME (h)}}$$

B TYPE

$$\text{AVERAGE VEHICLE SPEED (mph)} = \frac{\text{CUMULATIVE TRAVELED DISTANCE (mile)}}{\text{CUMULATIVE TRAVELED TIME (h)}}$$

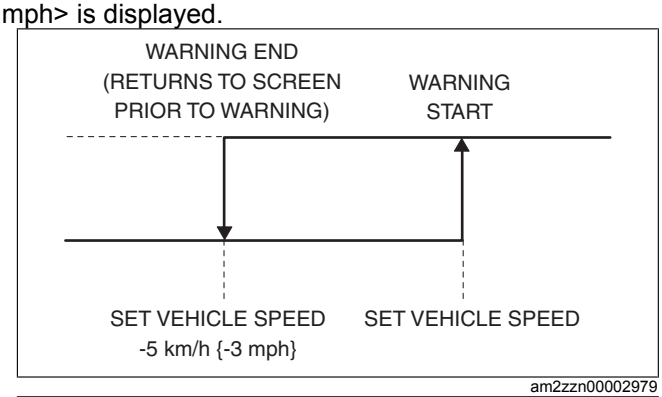
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- If the INFO button is pressed for **1.5 s or more** while the average vehicle speed is displayed, the average vehicle speed (cumulative traveled distance and cumulative traveled time data) is reset.
- During the **1 min period** after the data has been reset, <--- km/h> or <--- mph> is displayed.

Vehicle speed alarm function (With vehicle speed alarm)

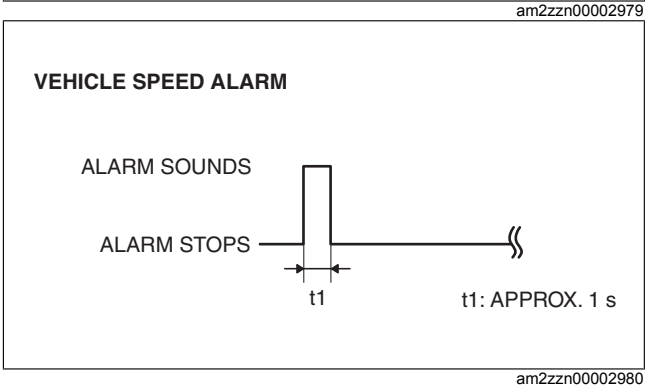
- The instrument cluster compares the current vehicle speed with the speed set by the user and if the current vehicle speed exceeds the set vehicle speed, the alarm is triggered.

- The set vehicle speed can be set in **10 km/h** or **10 mph** increments between **30 km/h—250 km/h** or **20 mph—150 mph**.
- If the vehicle speed alarm is not set, <--- km/h> or <--- mph> is displayed.
- The instrument cluster outputs a warning display and warning sound when the current vehicle speed is the same as the set vehicle speed.
- The instrument cluster switches the warning display back to the screen prior to the warning display when the current vehicle speed is **-5 km/h {-3 mph} or less** than the set vehicle speed.



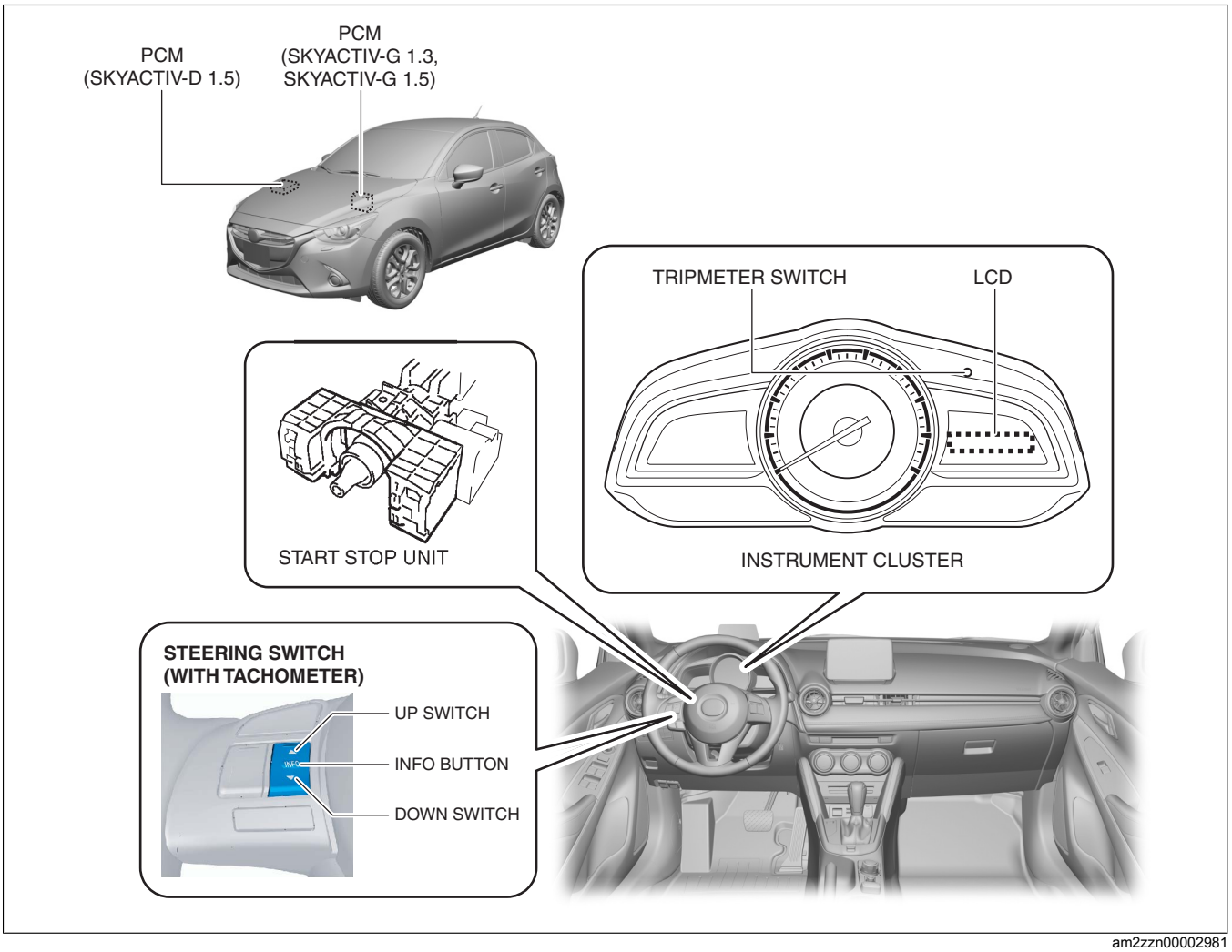
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- The alarm sound pattern is as shown in the figure.



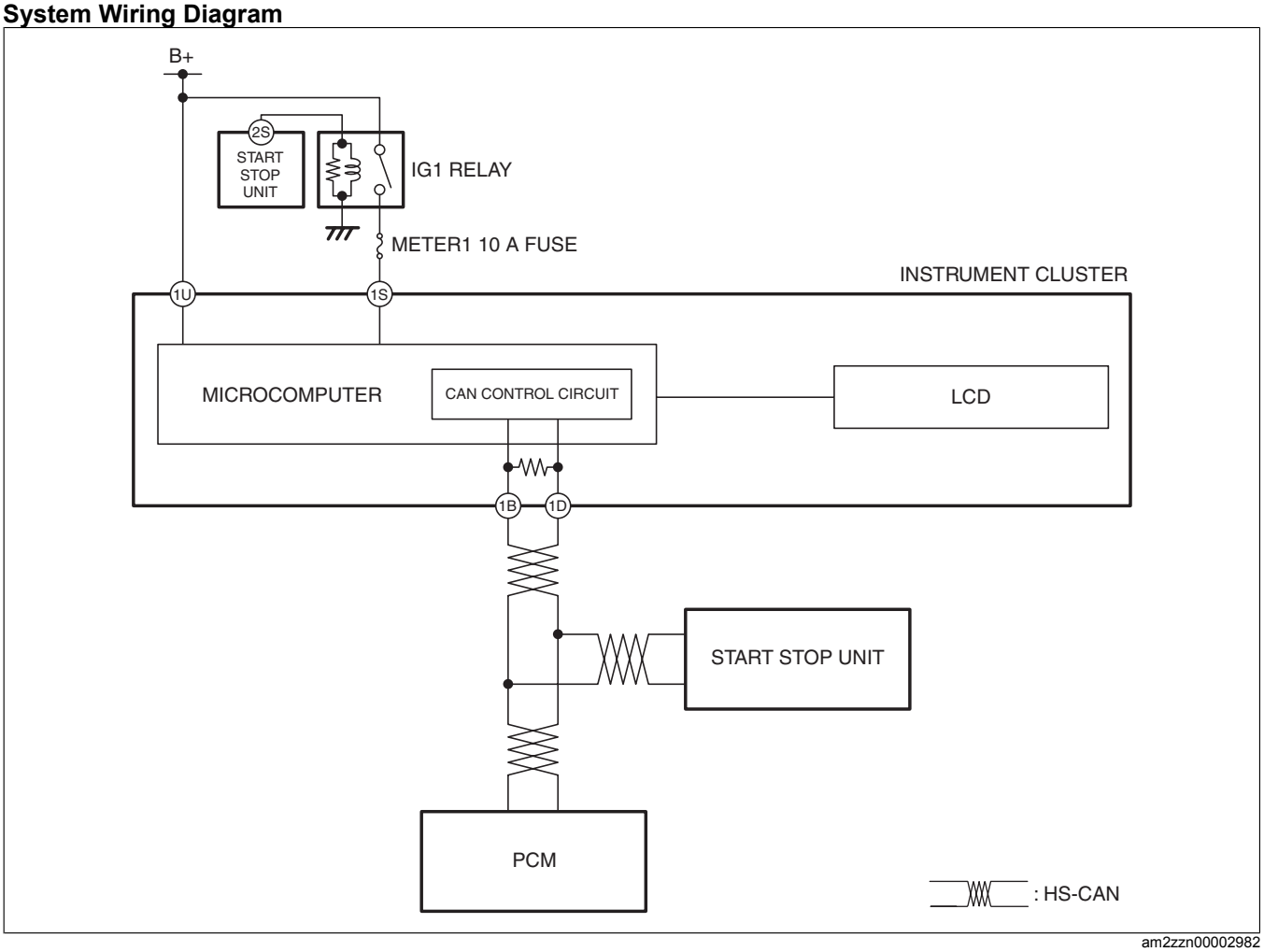
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Structural View



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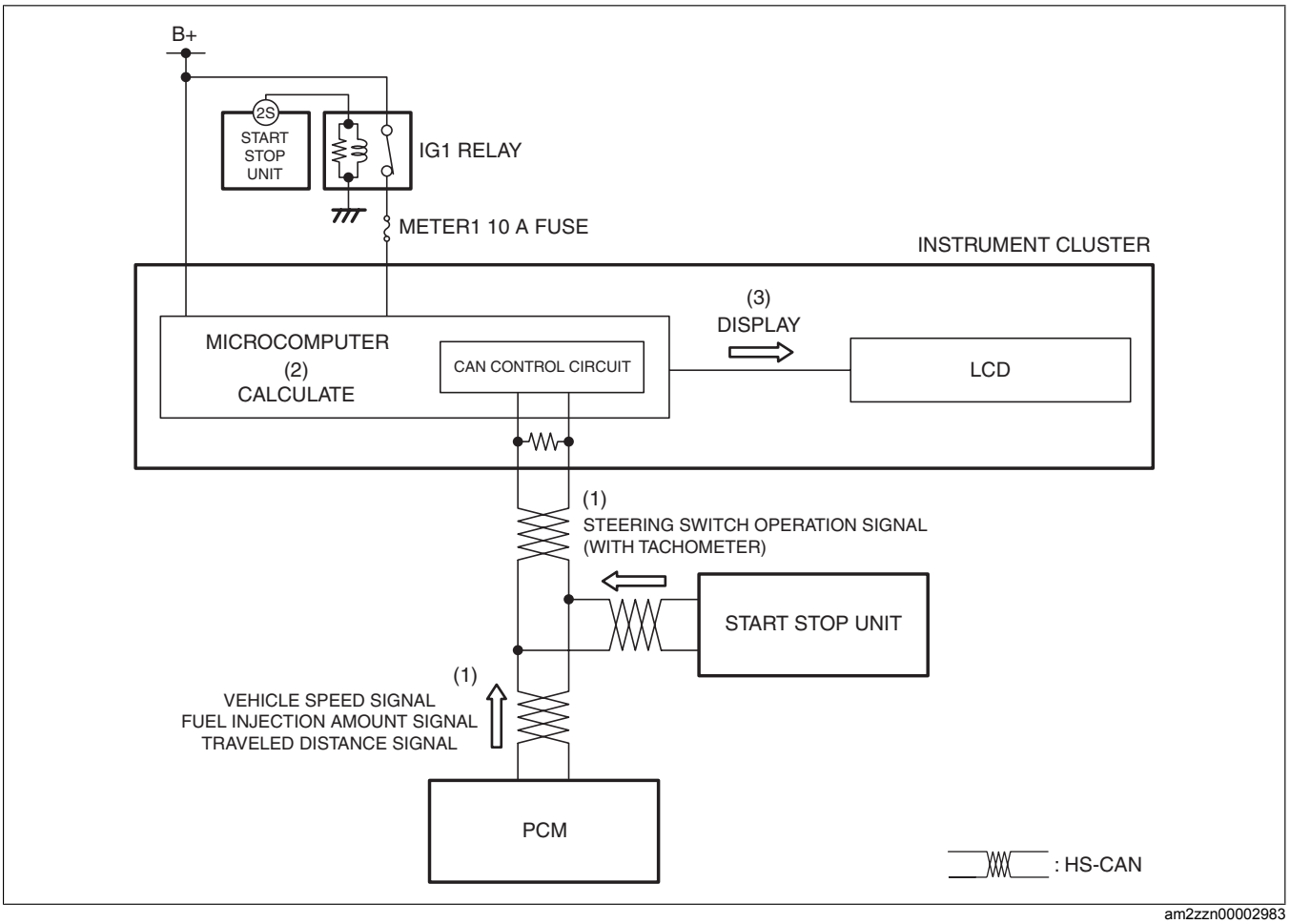
System Wiring Diagram



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Operation

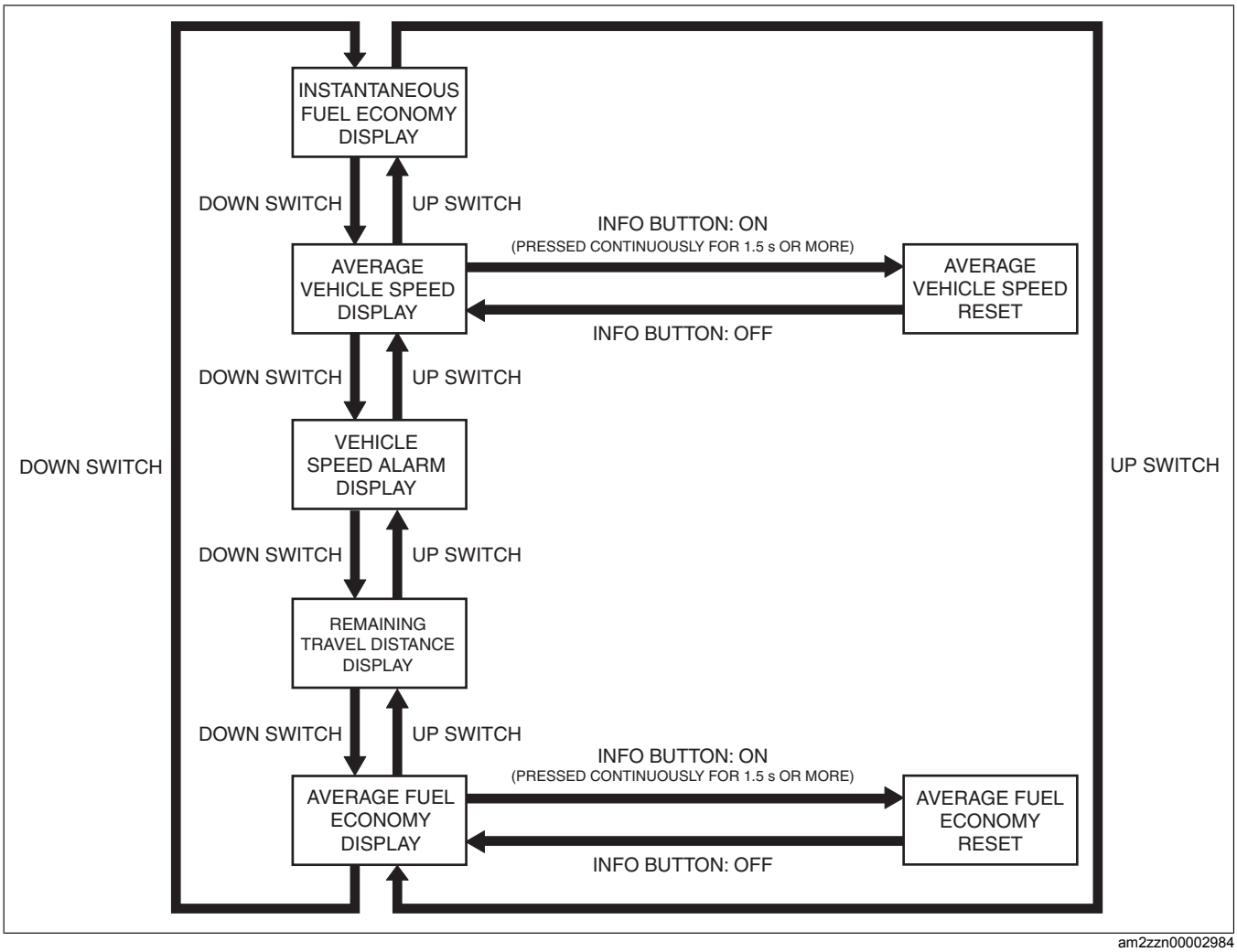
1. When the ignition is switched ON (engine on), the instrument cluster receives (1) the vehicle speed signal, fuel injection amount signal, and traveled distance signal from the PCM, and the steering switch operation signal (with tachometer) from the start stop unit via CAN communication.
2. The instrument cluster calculates (2) the instantaneous fuel economy and others based on the received data.
3. The instrument cluster displays (3) the calculated results in the LCD.



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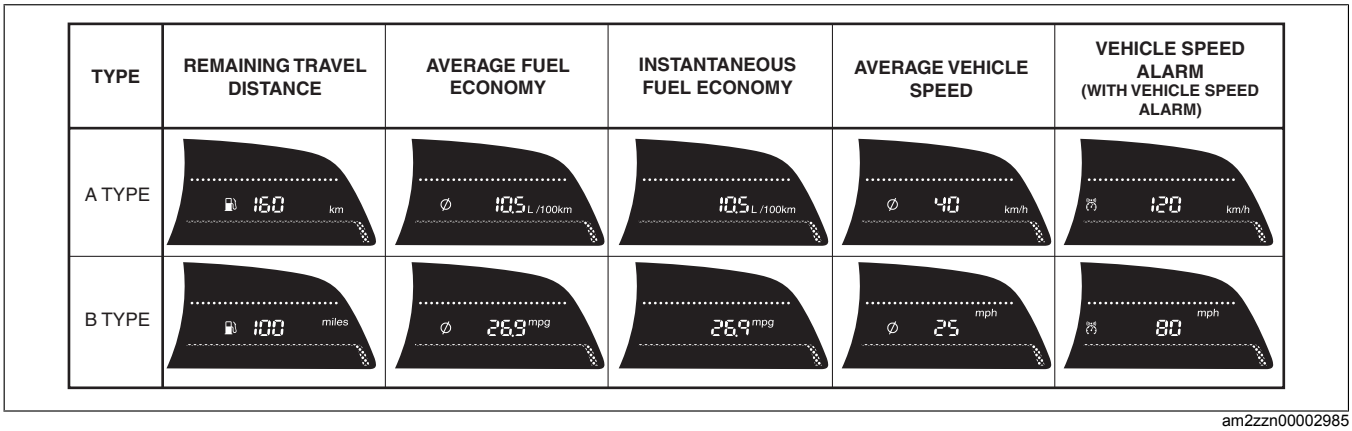
Trip computer display switching operation
Vehicles with tachometer

- The instrument cluster switches the trip computer display content as follows based on the steering switch operation signal from the start stop unit.



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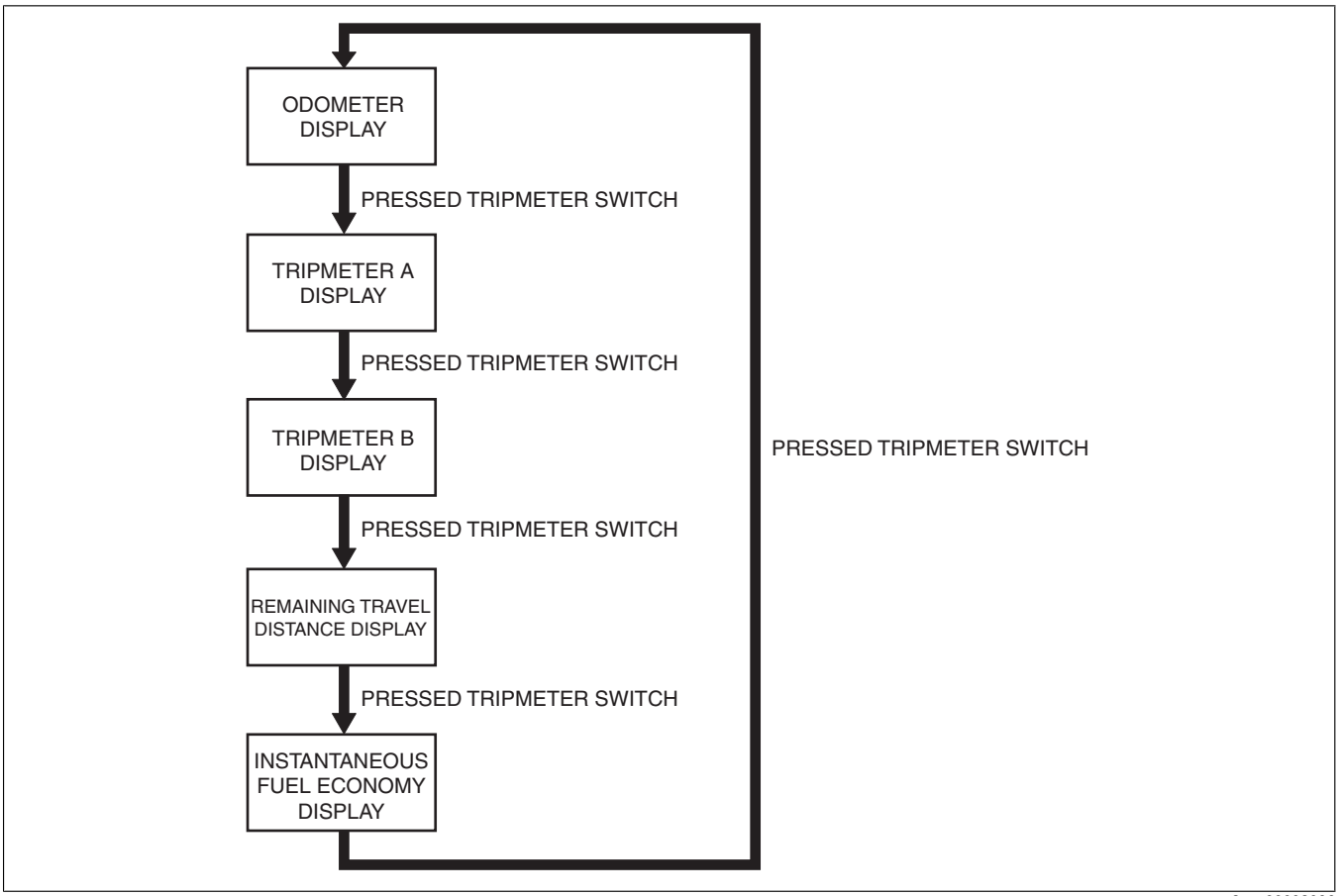
Display screen table



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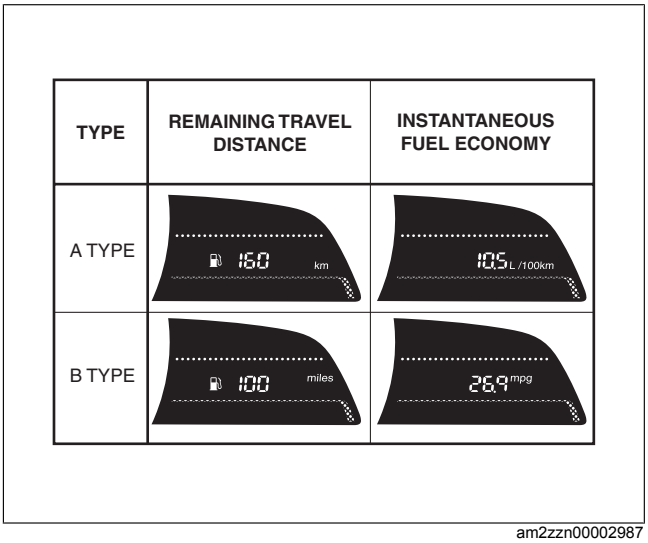
Vehicles without tachometer

- When the tripmeter switch is pressed, the instrument cluster switches the tripmeter display content as follows:



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Display screen table



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